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Set Up a Web App Using AWS and VS Code



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The screenshot shows the Visual Studio Code interface. On the left, the Explorer sidebar displays the project structure for 'NEXTWORK-WEB-PROJ...'. The structure includes a 'src/main' directory containing 'resources', 'webapp', and 'pom.xml'. The 'webapp' directory contains 'WEB-INF' (with 'web.xml') and 'index.jsp'. The 'index.jsp' file is selected and open in the editor. The editor shows the following HTML code:

```
src > main > webapp > index.jsp > ...
1  <html>
2
3  <body>
4
5  <h2>Hello {YOUR NAME}!</h2>
6
7  <p>This is my NextWork web application working!</p>
8
9  </body>
10
11 </html>
12
```



Introducing Today's Project!

I'll set up a web-app using EC2, VS code and dockers.

Key tools and concepts

Services i used today were EC2 with Keypairs, VS code. key concepts i learnt include, Linux, Maven, Java, Ec2, SSH and remote access.

Project reflection

I didnt expected the VS code use in this project.

It took me almost 50 minutes because i already had some of these things installed.

This project is part one of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project in 2 to 3 days.



Launching an EC2 instance

We launch an EC2 as computer for our web application.

I also enabled SSH

SSH i.e. Secure Shell is a protocol used to make sure only authorized users can access a remote server

Key pairs

Key pair is a security key to access the EC2.

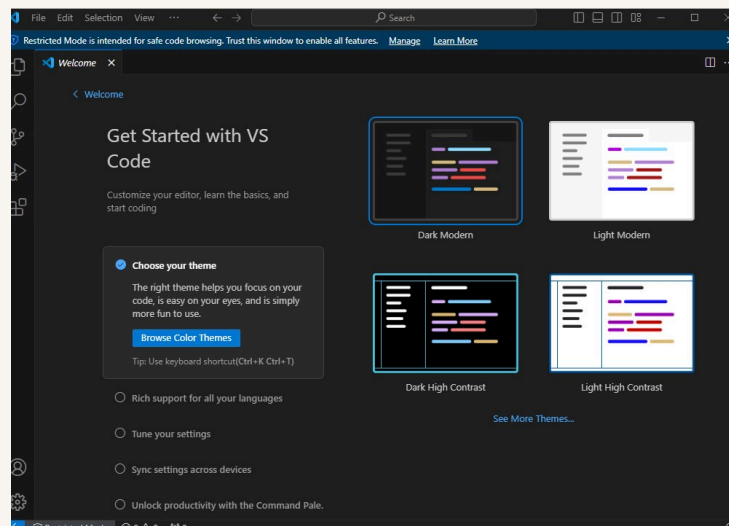
Once I set up my key pair, AWS automatically downloaded the key pair in .pem format.



Set up VS Code

Visual Studio Code, commonly referred to as VS Code, is an integrated development environment developed by Microsoft for Windows, Linux, macOS and web browsers.

I installed VS code to access my EC2 remotely.





My first terminal commands

This is the first command i ran `cd C:\Users\YourUserName\Desktop\DevOps`

I also updated my private key's permissions by running this command: `icaccls "nextwork-keypair.pem" /reset`
`icaccls "nextwork-keypair.pem" /grant:r "SayedMahmeer:R"`
`"USERNAME:R" icaccls "nextwork-keypair.pem" /inheritance:r`

```
PS C:\Users\SayedMahmeer\OneDrive\Desktop\DevOps> icaccls "nextwork-keypair.pem" /reset
>> icaccls "nextwork-keypair.pem" /grant:r "SayedMahmeer:R"
>> icaccls "nextwork-keypair.pem" /inheritance:r
>>
processed file: nextwork-keypair.pem
Successfully processed 1 files; Failed processing 0 files
processed file: nextwork-keypair.pem
Successfully processed 1 files; Failed processing 0 files
processed file: nextwork-keypair.pem
Successfully processed 1 files; Failed processing 0 files
PS C:\Users\SayedMahmeer\OneDrive\Desktop\DevOps> 
```




Maven & Java

Apache Maven is a tool that helps developers build and organize Java software projects

Maven is required in this project because this project is going to use a java-based programming language.

Java is a popular programming language used to build different types of applications, from mobile apps to large enterprise systems.

Java is required in this project because we are going to build a Java-based Application.



Create the Application

I generated java app using command : `mvn archetype:generate \ -
DgroupId=com.nextwork.app \ -DartifactId=nextwork-web-project \ -
DarchetypeArtifactId=maven-archetype-webapp \ -DinteractiveMode=false`

I installed SSH-remote to connect to my EC2.

Configuration details require complete path to our key pair, IPV4 DNs, and host name.

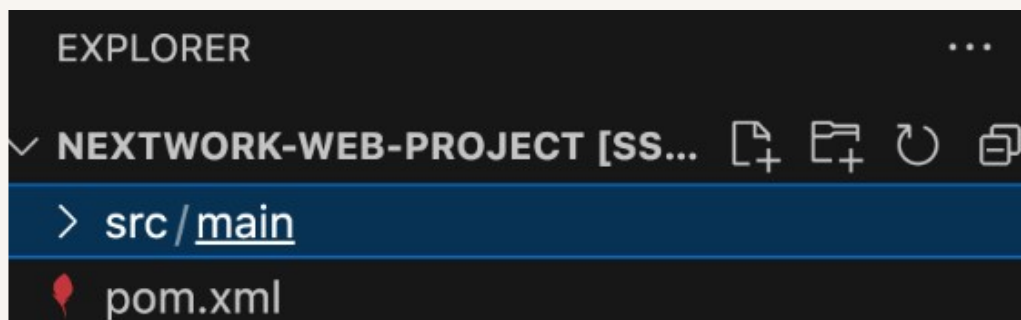
```
[INFO] Parameter: basedir, Value: /home/ec2-user
[INFO] Parameter: package, Value: com.nextwork.app
[INFO] Parameter: groupId, Value: com.nextwork.app
[INFO] Parameter: artifactId, Value: nextwork-web-project
[INFO] Parameter: packageName, Value: com.nextwork.app
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/ec2-user/nextwork-web-project
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 9.934 s
[INFO] Finished at: 2025-06-15T14:14:13Z
[INFO] Final Memory: 19M/84M
[INFO] -----
[ec2-user@ip-172-31-31-163 ~]$
[ec2-user@ip-172-31-31-163 ~]$
[ec2-user@ip-172-31-31-163 ~]$
```



Create the Application

Using VS Code's file explorer, I could see. src, web, index, and such files.

Two main folders created by Resources and Web-App

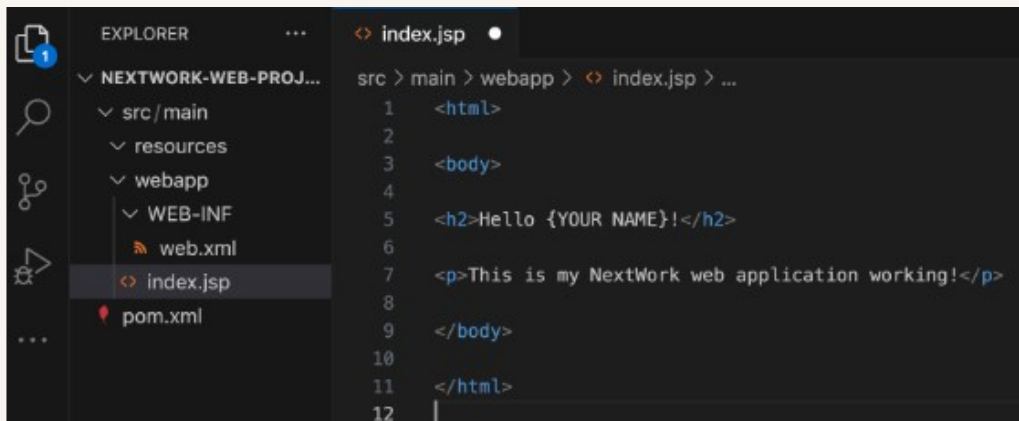




Using Remote - SSH

It is our file where we write the code.

i edited the file to modify with my name.



```
EXPLORER
  NEXTWORK-WEB-PROJ...
    src/main
      resources
      webapp
        WEB-INF
          web.xml
          index.jsp
        pom.xml

index.jsp
src > main > webapp > index.jsp > ...
1  <html>
2
3  <body>
4
5  <h2>Hello {YOUR NAME}!</h2>
6
7  <p>This is my NextWork web application working!</p>
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